J. ROBERT MANUEL

Software Quality Assurance (SQA) · Verification & Validation (V&V) Test Engineer

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Technical Interview Resume — Software QA / V&V Test Engineering

PROFESSIONAL SUMMARY

Software quality assurance and V&V engineer since 2011 in FDA Class I and Class II medical-device environments (Bayer / Medrad CT and MRI contrast injectors, Philips Respironics), FDA-regulated clinical-trial software, HIPAA-compliant healthcare applications, and embedded / IoT systems. Lead V&V SQA Engineer at Bayer and Lead SQA, Level IV at Omnicell; builds VMware ESXi test labs and VM templates for unattended automated testing.

CORE COMPETENCIES

Virtualization & Test Automation

- Built VMware ESXi virtual test environments and labs (Bayer; Software Design Solutions; Omnicell), including an internal ESXi lab from repurposed servers.
- Created VM templates for unattended automated testing alongside manual testing, supporting an SDLC with multiple live, resourced virtual machines.
- Managed Jira and Microsoft server farms for test environments.

Software Quality Assurance & Regulated Systems

- Verification & Validation (V&V) of FDA Class I and Class II medical devices and control software; full-SDLC SQA
- Regulatory frameworks: FDA QSR 21 CFR Part 820, ISO 13485, IEC 62304, ISO 14971, IEC 60601, ISO 9001; HIPAA
- Requirements analysis and traceability; test planning, protocol authoring and execution, dry runs, verification reports
- Functional, negative, stress, integration, UI, performance, and embedded / IoT testing; DTR / defect capture and resolution
- QA team leadership, SQA mentoring, and developer–QA "translation" in Agile / daily-scrum environments

SOFTWARE QUALITY ASSURANCE EXPERIENCE

Omnicell

Lead Software Quality Assurance Analyst — SQA Level IV

Omnicell is a provider of medication-management and pharmacy-automation solutions — robotics, smart devices, and software — for health systems and pharmacies.

- Built VMware virtual test environments and VM templates for unattended testing, as the SQA engineer on a small team with five developers, led by Derek.
- Worked with HP QTP and a Test Center environment, together with JIRA-style tracking and automation tooling.

Management Science Associates (MSA) — Pittsburgh, PA — 8/2018 – 2/2019

Software Quality Assurance Analyst, Life Sciences Division · Contract via Pinnakle Technologies

- Web application SQA for the Life Sciences Division, including UI manual testing of EditLife (editlife.msa.com).
- Tested a large-scale, enterprise-level organ-transplant application used across a hospital network, in a HIPAA-compliant environment.

Software Design Solutions (an Applied Visions company) — Murrysville, PA — 5/2017 – 4/2018

Senior Software Quality Assurance Analyst — Embedded Software & Hardware / IoT (Full-time)

- Embedded with a small development team as the QA/test engineer for bus, train, and ferry transit applications used to run public transit systems in the United States.

- Sole in-house SQA engineer for an internal development team of about 10 engineers working across multiple projects.
- Tested the upgrade of NASA satellite ground-control software (L3 / TTS program — name to confirm), migrated from legacy C++ during the relocation of the system from Arizona to Australia.
- Built an internal VMware ESXi virtualization lab from repurposed servers; created VM templates enabling unattended automated testing alongside manual testing, establishing an early SDLC with multiple live, resourced virtual machines.
- Performed software testing on the F-35 pilot helmet-mounted display, which projects flight and weapons data onto the pilot's visor.
- Performed SQA on embedded software and hardware (IoT) in an Agile environment — functional, negative, stress, integration, and regression testing.
- Created test plans and test cases, with a reporting structure in TestRail, even when requirements were unavailable or incomplete.
- Worked with Ed Kuzemchak, founder of Software Design Solutions.

Celtic Healthcare — Mars, PA **10/2016 – 4/2017**

Software Quality Assurance Engineer · Contract via Oxford Solutions, converted to full-time

- Tested web-based healthcare documentation and medical-coding software in an Agile team.
- UI and functional-integration testing; created and executed SQA tests tracked in JIRA.

Ciber — Pittsburgh, PA **3/2016 – 9/2016**

Senior Software Quality Assurance Engineer / Sr. QA Specialist (Full-time, regular)

- Tested a web-based Institutional Review Board (IRB) system supporting FDA-regulated clinical-trial software and document management.
- UI and functional-integration testing using Seapine TestTrack ALM; wrote and executed SQA test procedures.

Bayer Pharma Radiology & Informatics (formerly Medrad) — Indianola, PA **9/2014 – 2/2016**

Lead Verification and Validation Software Quality Assurance Engineer (Medical Device) · Contract via Premiere Technology Services (PTS), Walton Hills, OH

V&V of CT and MRI contrast-injector systems and control software in an FDA Class I / Class II regulated lab — including the Stellant / Certegra family (FDA Class II), used in hospitals worldwide for pediatric and premature-infant patients, with dual injectors timed between heartbeats.

- Worked on the Certegra MRI injector.
- Built and managed VMware ESXi virtual test environments, Jira, and Microsoft server farms.
- Performed CT and MRI injector V&V under FDA QSR 21 CFR Part 820 and ISO 13485; coordinated V&V plans, protocols, dry runs, test set-up validation, and verification reports.
- Conducted multi-injector physical pressure testing on approximately eight injectors at once, supported by background database scripts, frequently in GE MRI configurations.
- Captured discrepancies (DTRs) and drove them to resolution; performed automation testing with pre-scripted, low-level embedded C++-type code and tested HTML-based UI applications.
- Performed handwritten, hand-signed production release verification on a legacy device.
- Consulted with design teams on the day-in-the-life of the injector across CT, MR, PET, and interventional suites for NPD and LCE projects; guided use cases, task analyses, and workflows.
- Composed and verified controlled documentation before electronic signatures — three master books of roughly 500 pages each.

Philips Respironics — Monroeville, PA **9/2013 – 8/2014**

SQA Verification and Validation Test Engineer · Contract via Canon Recruiting Group, LLC, Santa Clarita, CA

- Design V&V on project teams developing medical devices — mechanical and electro-mechanical systems with embedded software and hardware for patient use.
- Wrote and executed detailed test plans against mechanical, electrical, and software requirements, with traceability across the SDLC.
- Tested legacy and new-hardware product designs; functional-requirements and performance testing.

K&L Gates, LLC — Pittsburgh, PA **9/2011 – 3/2013**

Software Quality Assurance Test Engineer & Application Support · Contract via Regency Group, Ltd., converted to full-time

- Led a start-up QA team for the firm's internal IT organization, implementing system image upgrades: Windows 7 images, remote SCCM deployments, and desktop installs.
- Used VMware vSphere server virtualization in core image testing; created and managed VMware ESXi environments.
- Tracked QA in Microsoft Team Foundation Server (TFS) with Visual Studio and Microsoft Test Manager; Level 3 support for WSUS.

TECHNICAL SKILLS & TOOLS

Test & ALM: HP Quality Center / ALM, HP QTP, Test Center, TestRail, Seapine TestTrack ALM, Microsoft Test Manager, Microsoft Team Foundation Server (TFS), Visual Studio

Requirements & PLM: IBM Rational DOORS, Enovia PLM, MatrixOne PLM, SAP document management, Visual SourceSafe

Tracking & collaboration: JIRA, Confluence, MKS Integrity Client

Data & back end: SQL Server Management Studio, MSSQL, SQLite, Oracle PL/SQL, Chrome REST client

Infrastructure: VMware vSphere / ESXi, SCCM, Active Directory, WSUS, Windows 7 imaging, Microsoft server farms, IT networking

ADDITIONAL IT EXPERIENCE

Computer Housecall — Pittsburgh, PA **2/1996 – 5/2000**

PC Field Technician · Help Desk — concurrent with my church music post

EDUCATION & TRAINING

Community College of Allegheny County (CCAC) — Pittsburgh, PA **1999**

Certificate in Information Technology — IT Networking

Northwestern University — Evanston, IL **1984 – 1988**

Bachelor of Music Education — student-teaching placements in three public school districts near Evanston

University of Miami — Coral Gables, FL **1996**

Master of Music, Choral Conducting — Full Teaching Assistantship (coursework 1988–1990; thesis and degree 1996)

CERTIFICATIONS & CREDENTIALS

- Regulated medical-device certifications: FDA 21 CFR 820 / ISO 13485 (quality systems); IEC 62304 (software life cycle); ISO 14971 and IEC 60601 (risk management; electrical safety).

REFERENCES

Professional and musical references are available on request.